EXP 20: Implement Future Sales Prediction using a suitable machine learning algorithm

CODING:

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# Future Sales Prediction using Linear Regression


# Historical sales data

months = [1, 2, 3, 4, 5, 6]

sales = [10000, 12000, 14000, 16000, 18000, 20000]


n = len(months)


# Calculate mean

mean_x = sum(months) / n

mean_y = sum(sales) / n


# Calculate slope

numerator = 0

denominator = 0


for i in range(n):

  numerator += (months[i] - mean_x) * (sales[i] - mean_y)

  denominator += (months[i] - mean_x) ** 2


slope = numerator / denominator


# Calculate intercept

intercept = mean_y - slope * mean_x
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# Predict future sales

future_month = 7

predicted_sales = slope * future_month + intercept

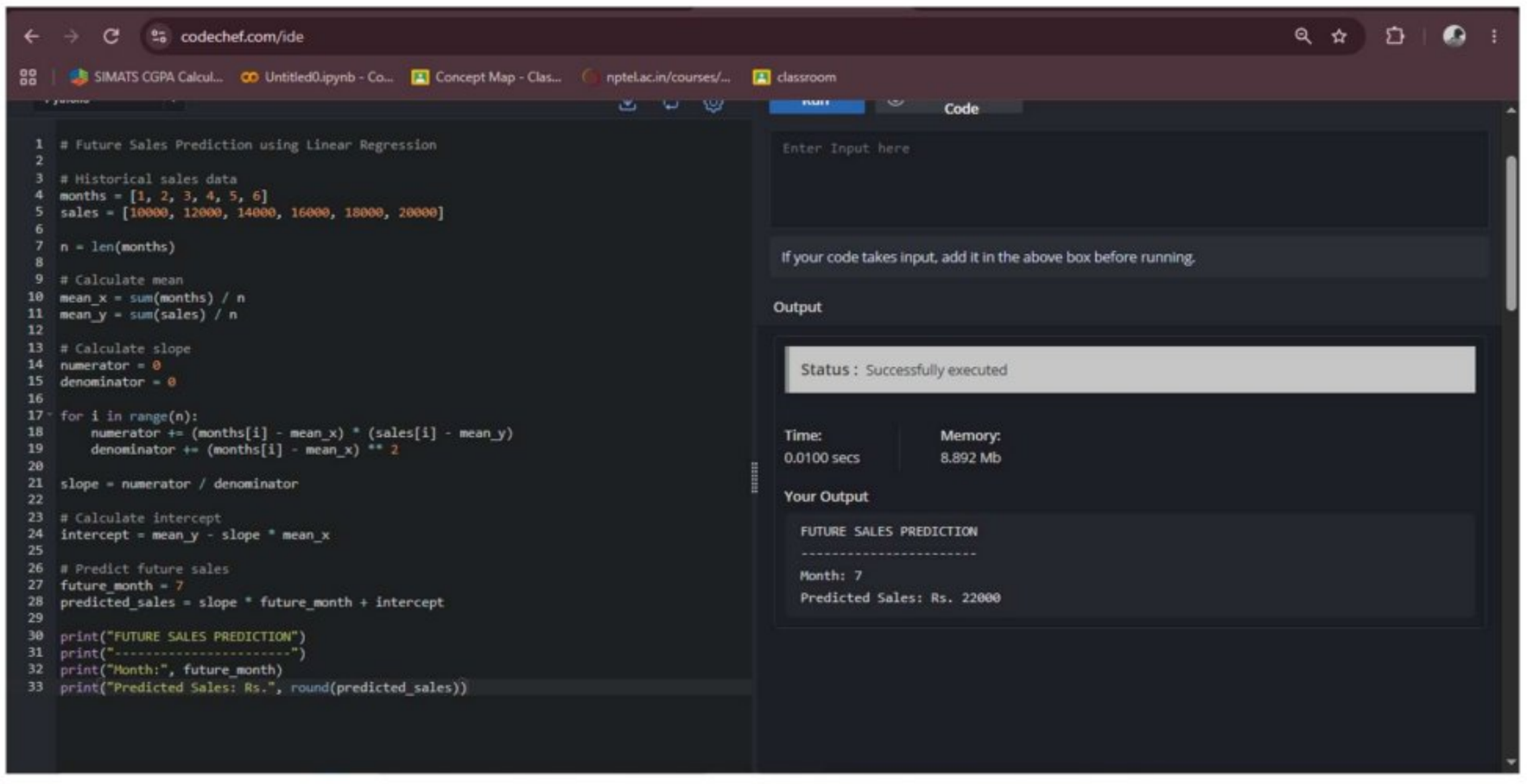

print("FUTURE SALES PREDICTION")

print("----------------------")

print("Month:", future_month)

print("Predicted Sales: Rs.", round(predicted_sales))
```

OUTPUT:



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